

Monthly Risk and Contribution to Risk Methodology

1 Introduction

To analyze the **monthly risk** and how each stock contributes to that risk, we can break down the process into three parts:

1. Monthly Risk of the Index: Measures the overall risk of the index at a monthly frequency.
2. Monthly Contribution to Risk: Measures how much each stock contributes to the risk of the index at a monthly level.
3. Monthly Delta Risk: Captures the change in the contribution to risk for each stock month-over-month.

This methodology does not rely on covariance matrices, standard deviation, or complex statistical measures, but instead focuses on simple return-based calculations.

2 1. Monthly Risk of the Index

The **monthly risk** of the index is calculated as the absolute change in the index return from one month to the next. This serves as a basic measure of risk, capturing the fluctuations in the index.

For month t , the risk of the index is given by:

$$\text{Risk}_{\text{Index},t} = |r_{\text{Index},t} - r_{\text{Index},t-1}|$$

where:

- $r_{\text{Index},t}$ is the return of the index for month t ,
- $r_{\text{Index},t-1}$ is the return of the index for the previous month.

3 2. Monthly Contribution to Risk by Each Stock

The **contribution to risk** of each stock is calculated by comparing the stock's return to the index return for the same month. This gives us an idea of how much each stock's fluctuations contribute to the overall risk of the index.

For stock i in month t , the **contribution to risk** is:

$$\text{Contribution to Risk}_{i,t} = |r_{i,t} - r_{\text{Index},t}|$$

where:

- $r_{i,t}$ is the return of stock i for month t ,
- $r_{\text{Index},t}$ is the return of the index for month t .

4 3. Monthly Delta Risk

Delta Risk captures the **change** in the contribution to risk from one month to the next. This allows you to see how the individual stocks' contributions to the index risk are evolving over time.

For stock i in month t , the **delta risk** is:

$$\Delta\text{Risk}_{i,t} = |r_{i,t} - r_{\text{Index},t}| - |r_{i,t-1} - r_{\text{Index},t-1}|$$

where:

- $r_{i,t}$ is the return of stock i for month t ,
- $r_{\text{Index},t}$ is the return of the index for month t ,
- The first term represents the contribution to risk for month t , and the second term is the contribution to risk for the previous month. The absolute difference between them shows the change (delta risk).